# Prompt Engineering

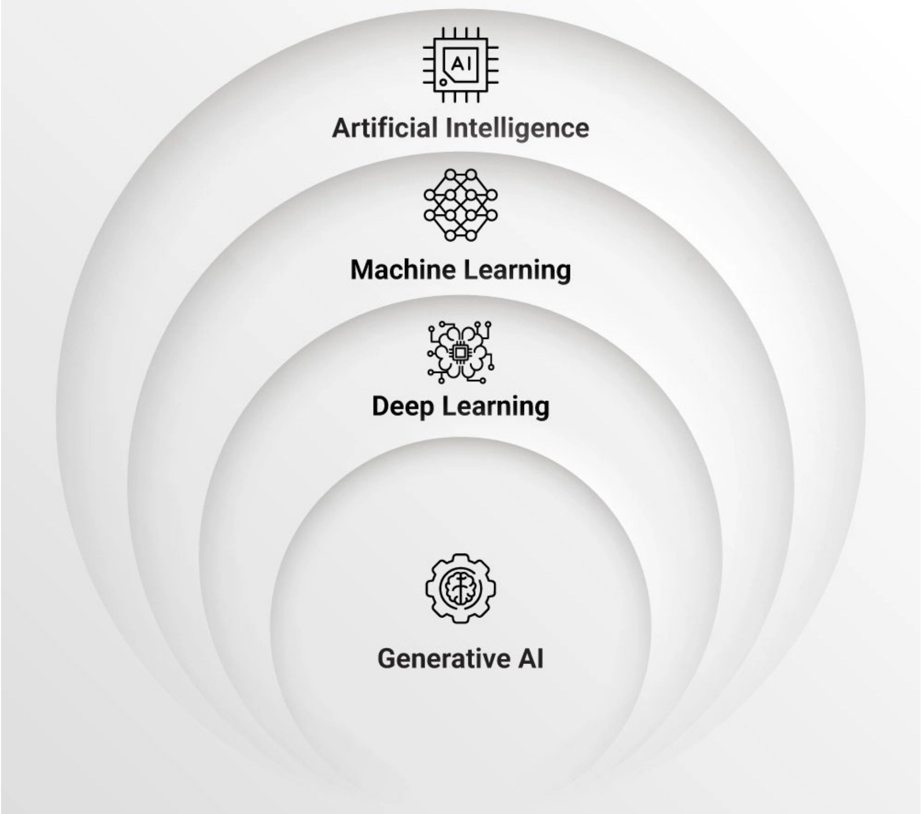
## 1:  AI Basics (Optional)

# Introduction

- This module provides some background on
  - Large Language Models
  - How LLM Prompts Work

# Defining AI

- AI includes a number of different fields of study
  - The current focus of AI is on machine learning, which includes LLMs and GAI
- There are other areas of AI that are not machine learning, for example
  - Cognitive computing
  - Artificial life and swarm intelligence
  - Evolutionary computing
  - Expert systems



# Genetic and Evolutionary Computation

- Uses algorithms modelled after natural selection and evolution
  - Often used to solve problems that are too computationally expensive to solve otherwise
  - For example, PostgreSQL uses genetic computation to help optimize queries
- These are part of AI that focuses on developing optimal constraint based problems

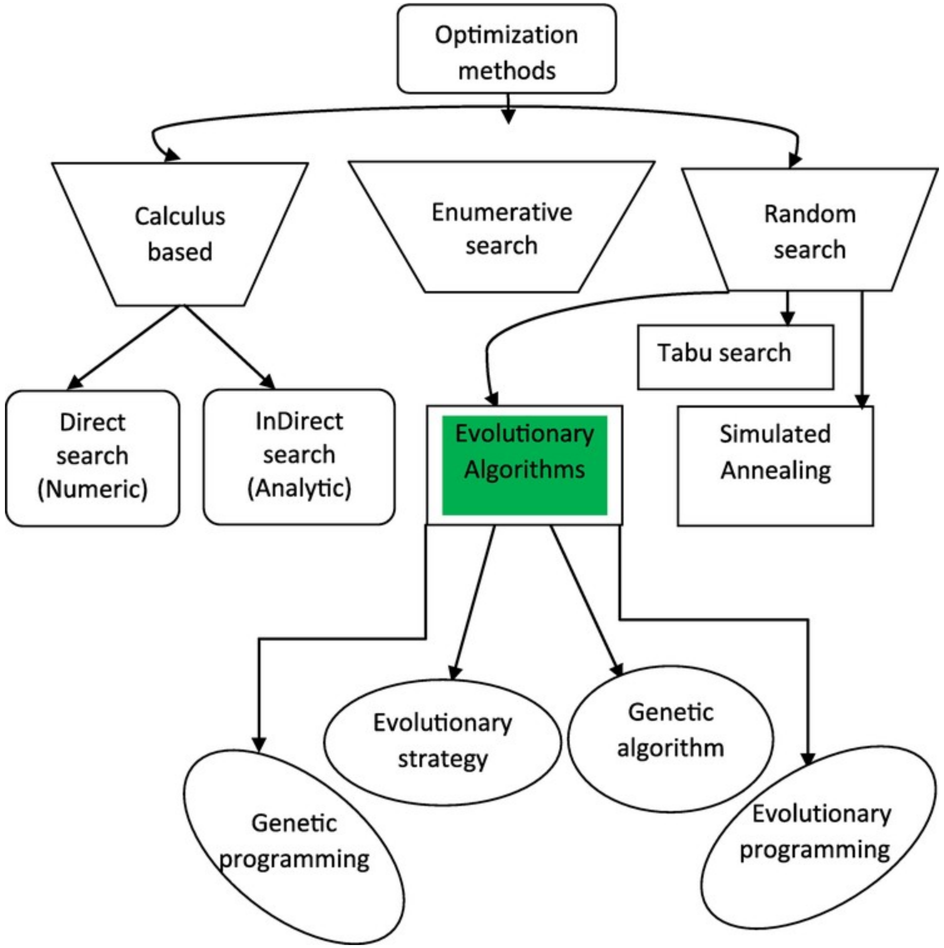



Image Credit: https://www.intechopen.com/chapters/81369

# AGI and ANI

- Artificial general intelligence (AGI)
  - AI that possesses human-like intelligence and is capable of understanding and learning any intellectual task that a human being can
    - Often referred to as “strong AI” or “full AI”
    - Able to perform a wide range of tasks and adapt to new situations without being explicitly programmed to do so
    - Beyond any current capability we have
- Artificial narrow intelligence (ANI)
  - AI systems that are designed for specific tasks or functions.
    - Specialized and limited in scope
    - Focusing on solving particular problems or performing specific functions within a narrow domain

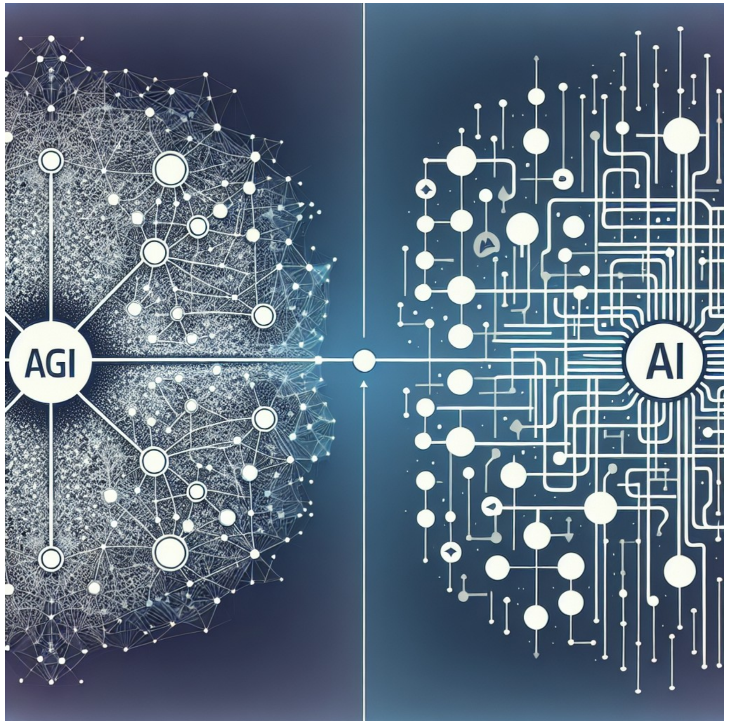
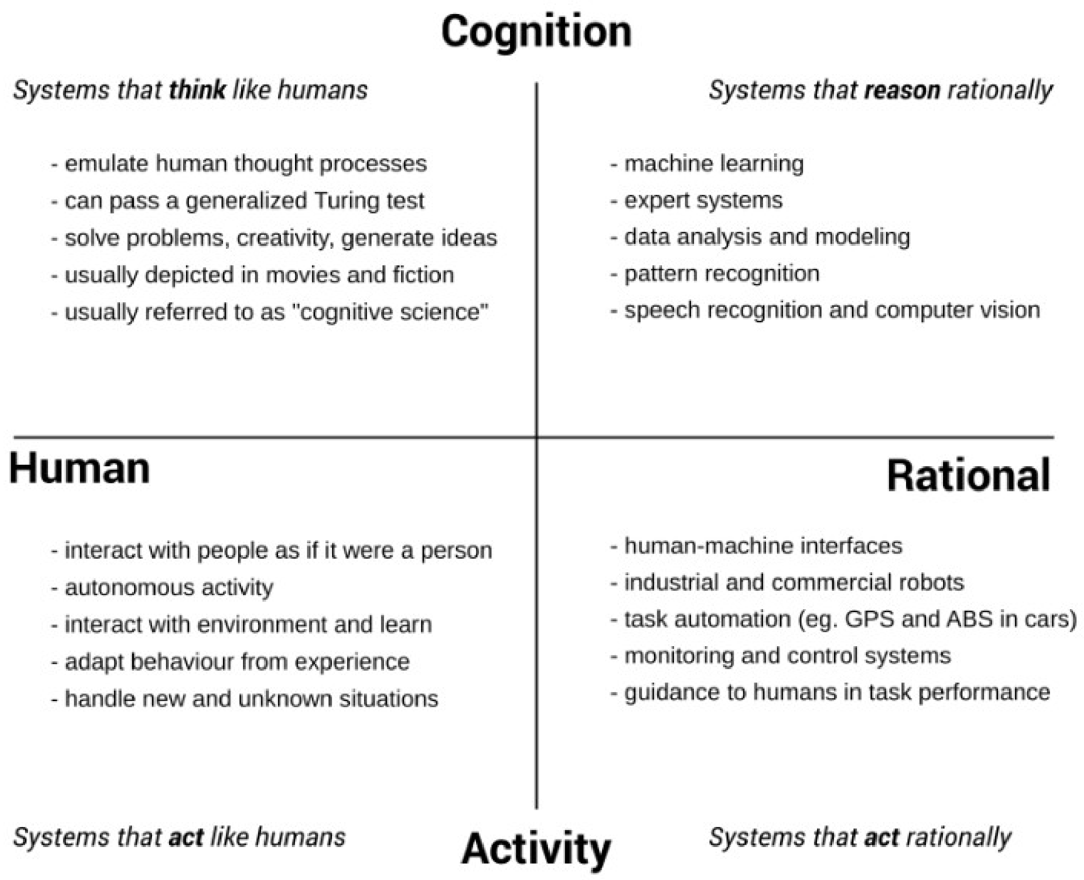



Image Credit: https://adamwalsworth.com/understanding-the-difference-between-agi-and-narrow-ai-what-sets-them-apart/

# The Norvig Matrix

- Provides a general classification of the different types of AI based on their goals
  - The "Human" column represents the goals of AGI, which are currently unattainable
    - We do not understand human cognition and behaviour enough to simulate it
    - Posited that we have to "grow" AGI by raising it like a child
  - The "Rational" column represents what we currently do in AI
    - Systems that reason rationally produce results that are acceptable to humans, although the techniques used are not the same as what people use
    - Systems that act rationally perform tasks whose outcomes are acceptable to humans but do not use the same processes as humans to execute the behaviour



# Classical Machine Learning

- Classical ML refers to statistical models created using different algorithms
  - These are drawn from other fields like statistics and data analysis
  - ML introduced techniques to create these models using “training” algorithms that worked on large data sets
    - Eg. The use of gradient descent to automatically converge to an answer
    - Eg. Measures to test how accurate and reliable the model was at making predictions
- Generally applied to a single problem
  - Spam detection in email
  - Tumour detection in medicine

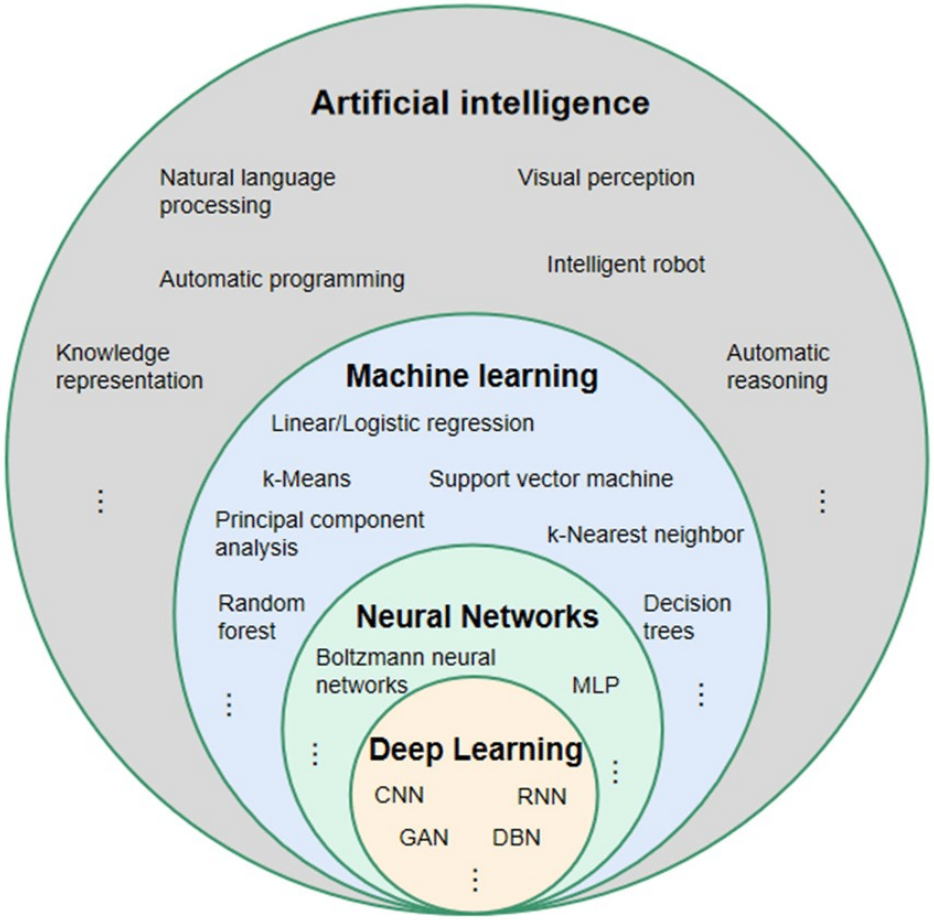
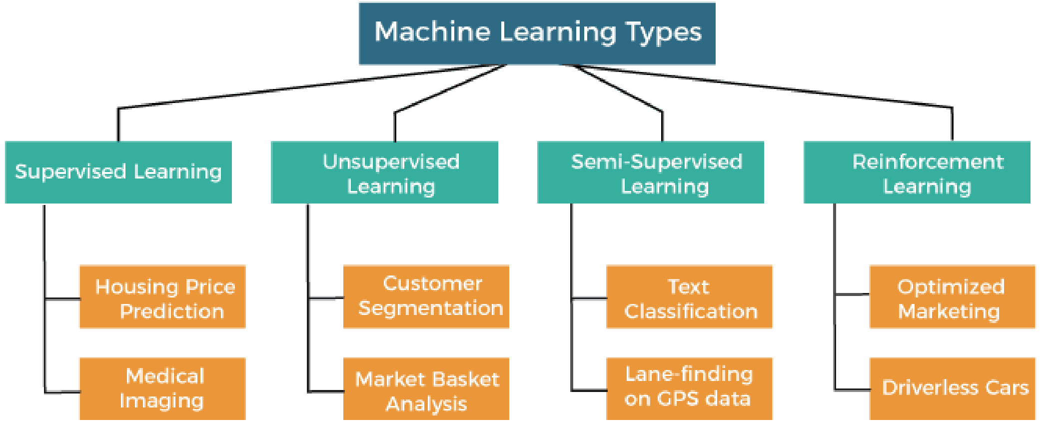



Image Credit: https://www.mdpi.com/2075-4418/11/9/1523

# Classical Machine Learning

- Three main classical types
  - Supervised learning
  - Unsupervised learning
  - Reinforcement learning
- Semi-supervised learning
  - Later addition to the classical types that came out of deep learning
- Used for specific problems
  - Models can be developed with small data sets and limited computation
  - Much less compute intensive than neural networks

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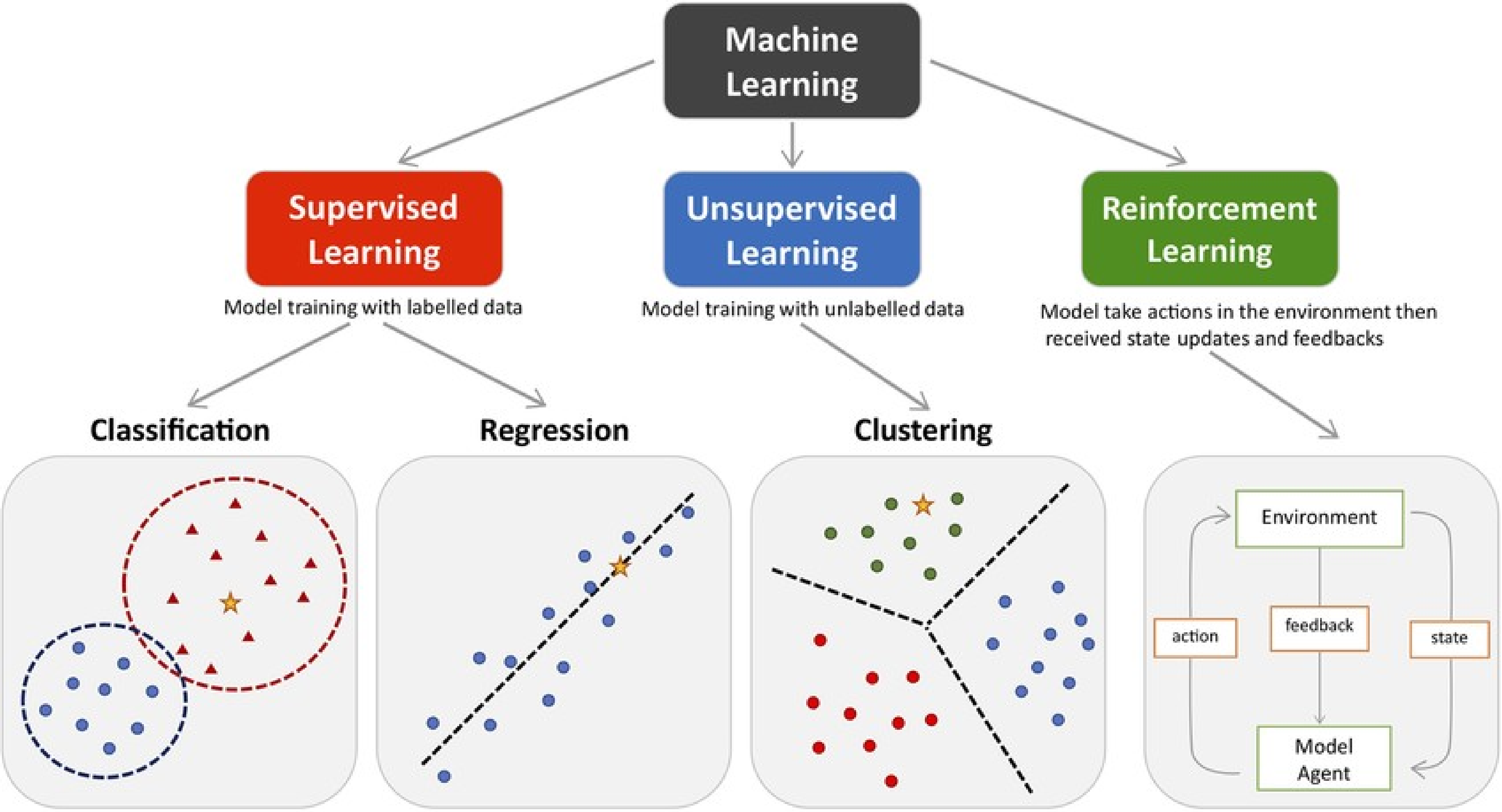
# Classical Machine Learning



# Neural Networks

- Artificial neural nets (ANN) are at the core of Deep Learning
  - Powerful, scalable and can solve complex problems like classifying billions of images
  - ANNs inspired by neurons in human brain
  - Original theoretical formulation in the 1940s
- First working models in the 1960s
  - Very limited in power and could not simulate actual neurons
  - But early work provided an architectural model for deep learning
  - No longer intended to simulate human neurons

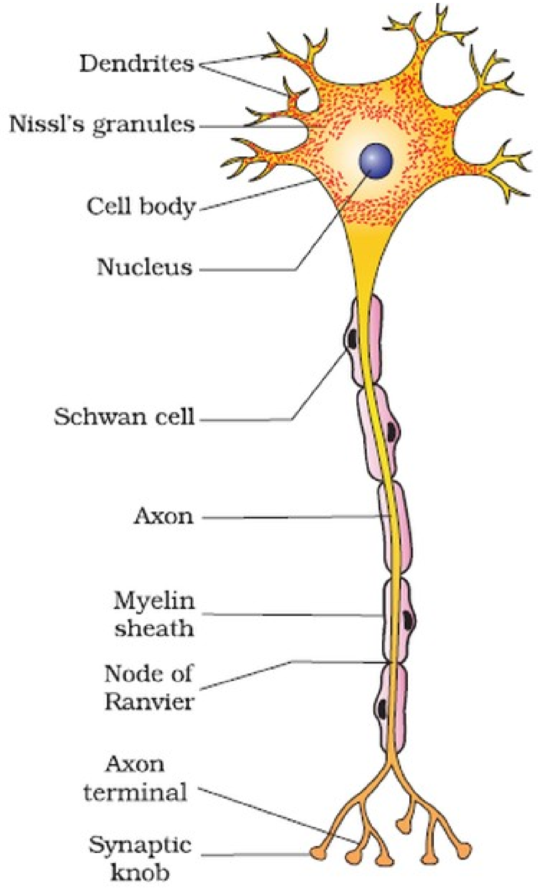



Image Credit: https://biology4isc.weebly.com/nervous-system.html

# Deep Learning

- Refers to the algorithms that execute in a neural net design
- A neural net is a software configuration
  - Nodes that represent abstraction based on the architecture of neurons and synapses
- Deep learning is the algorithm that produces AI models
  - Deep learning algorithms run on neural net architectures
  - Individual neurons are traditionally called "perceptrons"

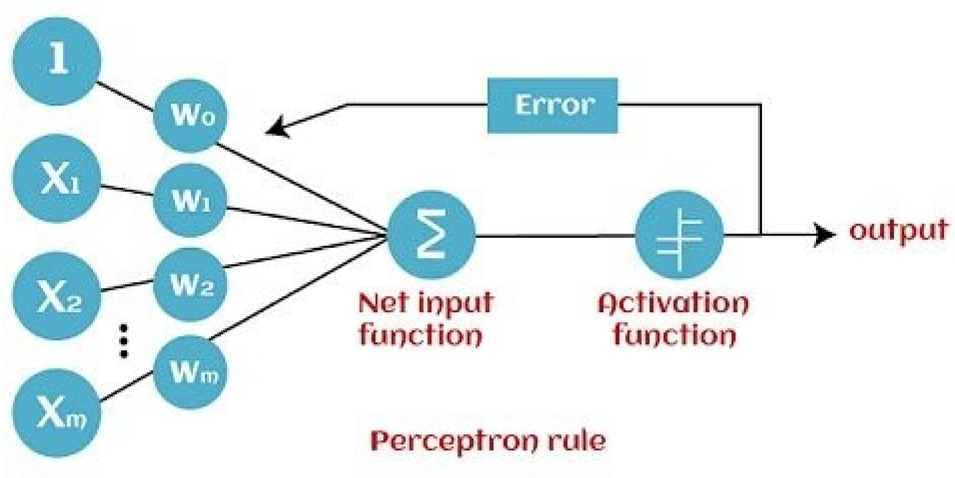



Image Credit: https://www.researchgate.net/figure/Working-diagram-of-the-perceptron-model-of-ANN_fig1_378886421

# Feed Forward Nets

- Have are multiple layers
  - Each layer has many neurons
  - Each neuron is connected to neurons on previous layer
- Information flows through in one direction - no feedback loop
  - Made up of input, output and hidden layers
  - Nets with more than one hidden layer are called deep neural nets
- Very difficult to do any form of deep learning on a feed forward net
  - No error correction feedback possible

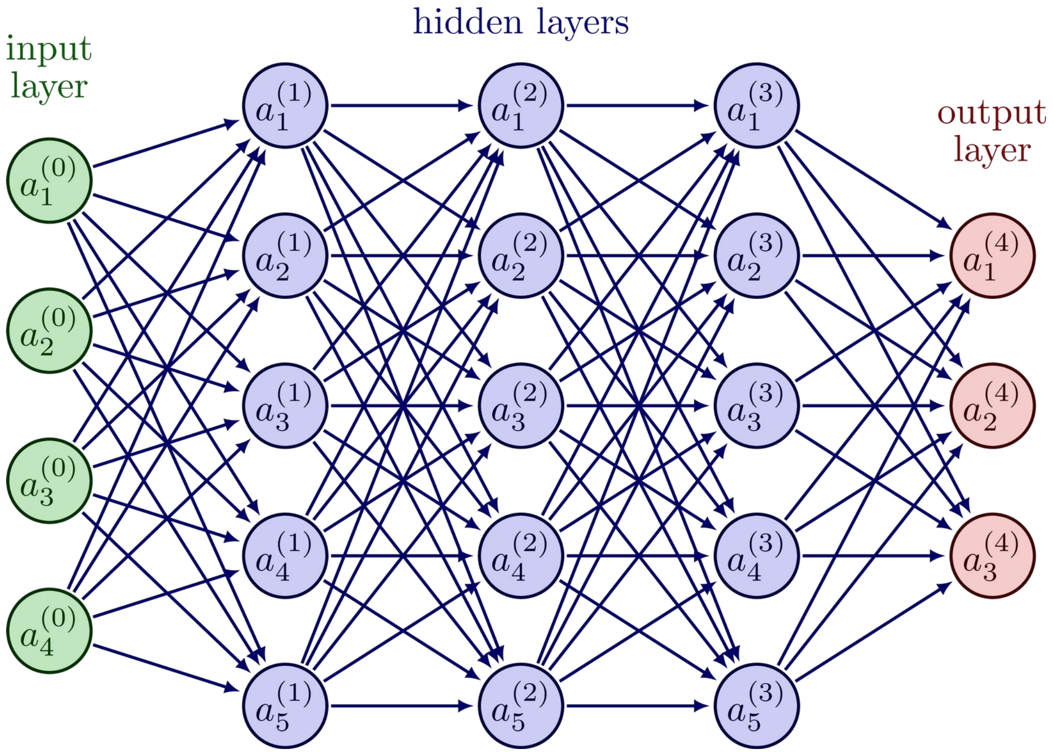



Image Credit: https://tikz.net/neural_networks/

# Deep Learning with Back Propagation

- Back propagation is the basis for training neural nets
  - A result is computed
  - The output is compared to the right answer
  - The last layer's inputs are "tweaked" to get it closer to the right answer
  - Then the previous layer is "tweaked" to get it closer to giving the last layer its tweaked inputs
  - This continues from back to front until all the layers have been tweaked.
- This is a very computationally expensive process and very time consuming

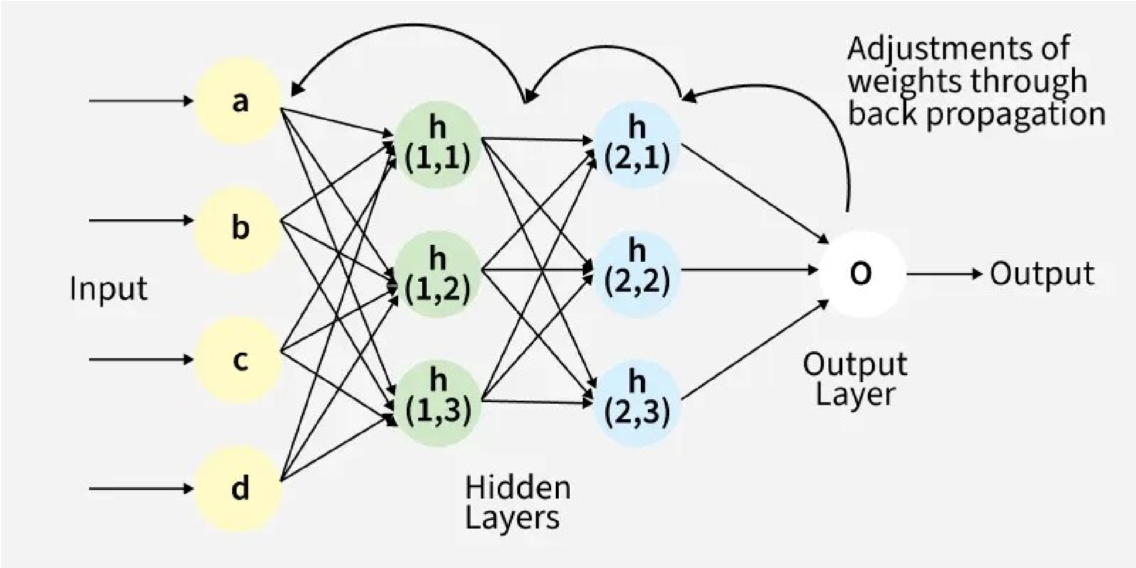



Image Credit:https://www.geeksforgeeks.org/machine-learning/backpropagation-in-neural-network/

# Convolutional NN

- CNNs were developed for image processing
  - Imagine a small patch being slid across the input image
  - This sliding is called convolving
  - Similar to a flashlight moving from the top left end progressively scanning the entire image
  - This patch is called the filter/kernel and the area under the filter is the receptive field
  - The idea is to detect local features in a smaller section of the input space, section by section, to eventually cover the entire image
- CNNs led to generative AI image processing

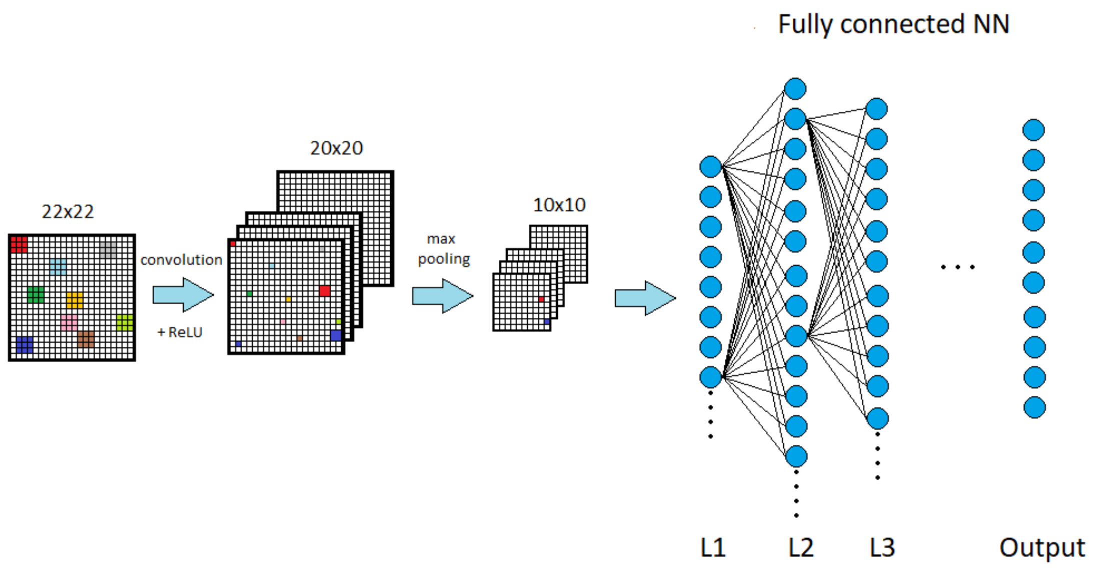



Image Credit:https://kim.hfg-karlsruhe.de/neural-network-chart/

# Problems with Text

- Text processing is more difficult that image processing
  - To understand text, the system has to remember what it read previously
  - It has to remember the context of the discourse
  - Context is needed for
    - Disambiguation of word meanings (was the screwdriver a drink or a tool?)
    - Anaphora resolution (who does “he” refer to?)
    - Basic understanding of what the text is about

Bob had always hated Dave. He always had.

One day, he was drinking at his favourite bar when he walked in. Furious that he would show up at his favourite bar, he threw a screwdriver in his face.

Dave wasn’t hurt so he declined to press charges.

# NN with Memory

- Early attempts at text processing used some form of memory to remember context
  - Recurrent NNs used a feedback loop to feed previous information back into the net
  - LSTM tried to refine this process by adding a memory module that would “forget” older information
- This led to techniques for allowing NNs to “pay attention” to what was important in context

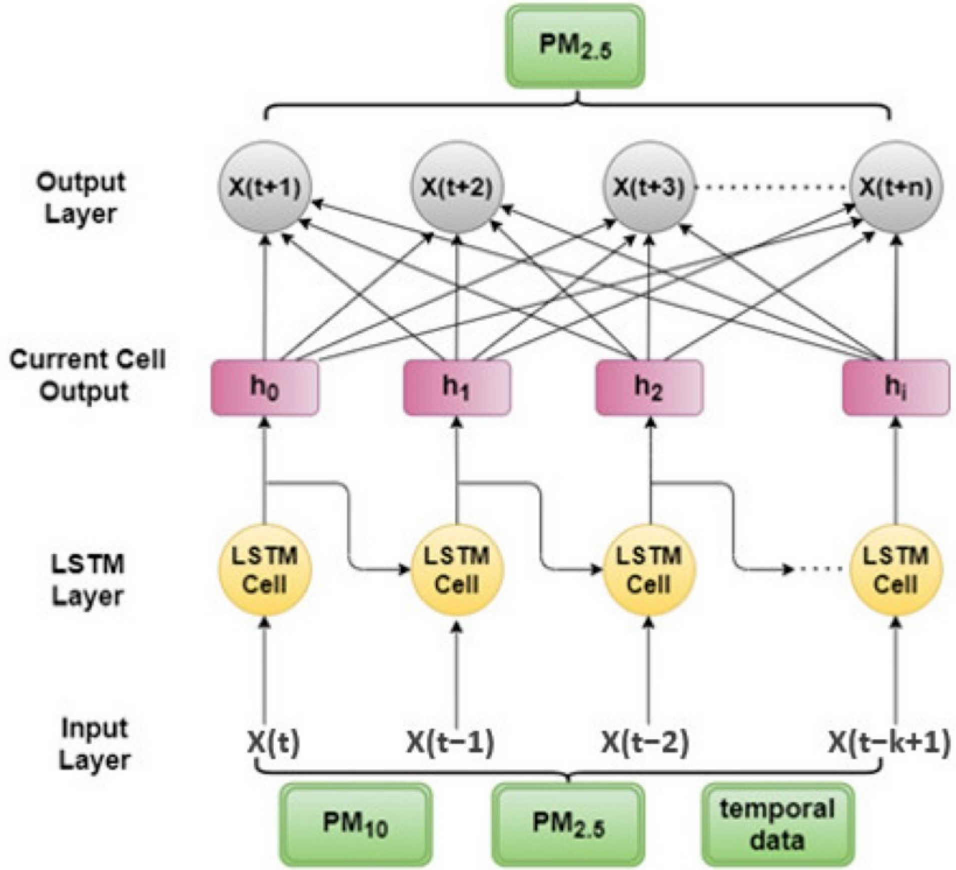



Image Credit: https://www.researchgate.net/figure/Long-short-term-memory-networks-LSTM-architecture_fig3_353082429

# Attention

- When you read, not every word has the same importance
  - Focus is on certain key words that carry meaning
    - “Although he was tired, John finished the report before the deadline.”
    - When trying to understand who finished the report, more attention is paid  to “John” and “finished” and less to filler words like “although” or “the”
  - Attention mechanisms help the model focus more on the important words when making a decision or prediction
    - “Not all words matter equally, so pay more attention to these ones that do matter”

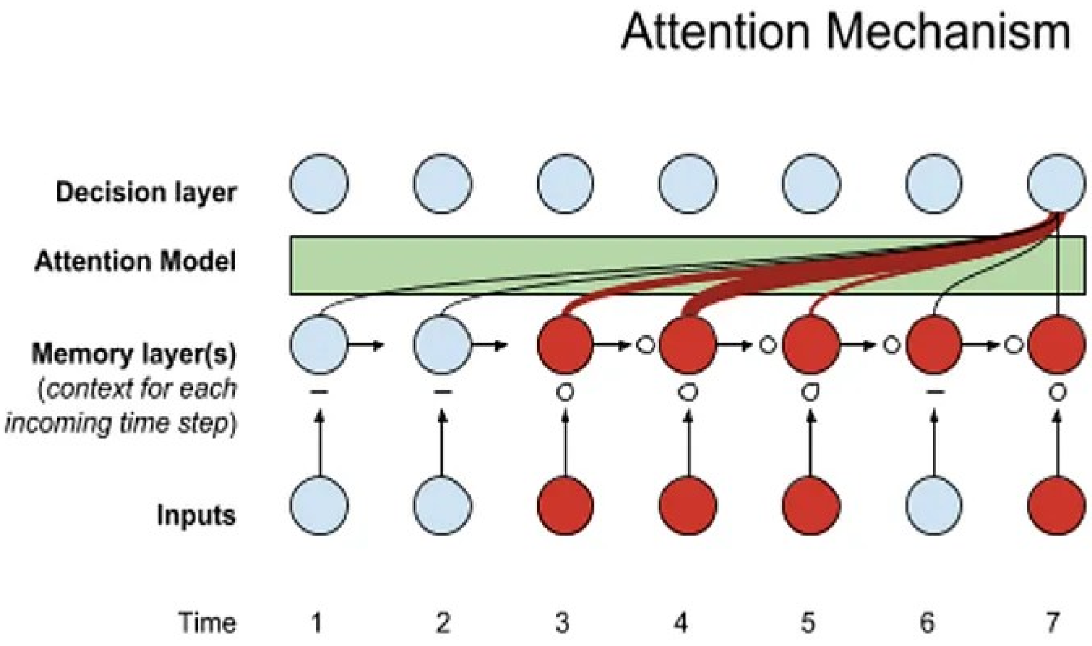
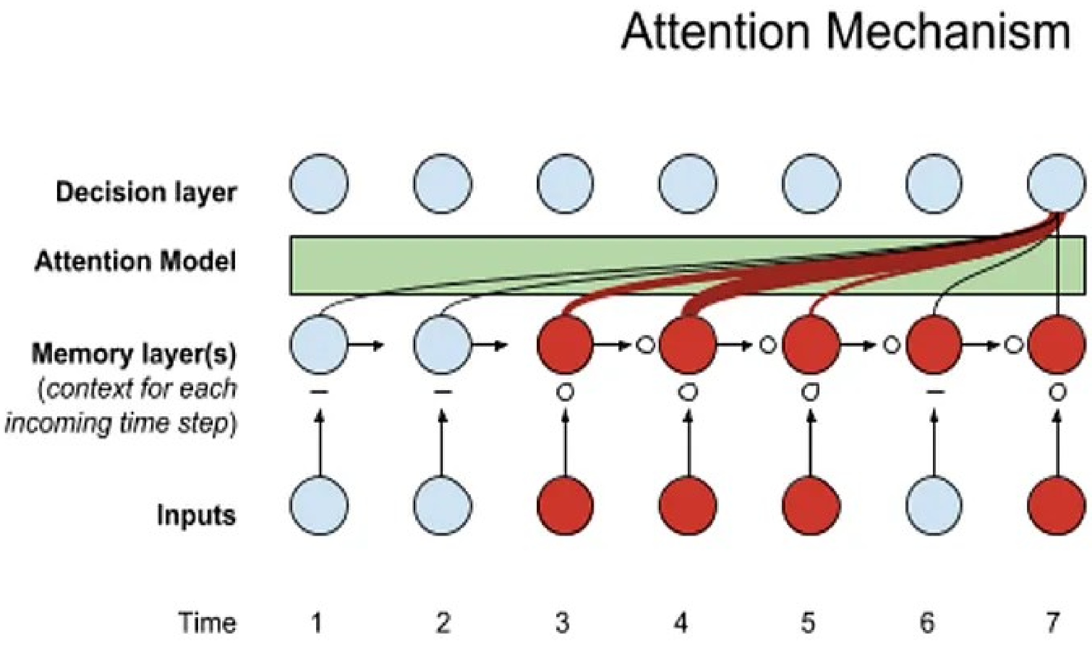



Image Credit: https://medium.com/@paruchurisaiteja15/attention-networks-3ab23176bb88

# Attention

- Attention requires analyzing a sentence and its context at the same time looking for patters of occurrence
- And then developing relationships in the model to reflect the patterns in the text

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# Transformers

- Transformers read an entire sentence, or paragraph, at once
  - They can figure out which words relate to each other, no matter where they appear
- “The bank approved the loan because the customer had excellent credit”
  - A transformer
    - Understands that “approved” is connected to “bank”
    - Connects “excellent credit” with the reason for approval
    - Doesn’t get confused if you rearrange the sentence slightly
    - Uses attention to weigh the importance of each word in relation to others

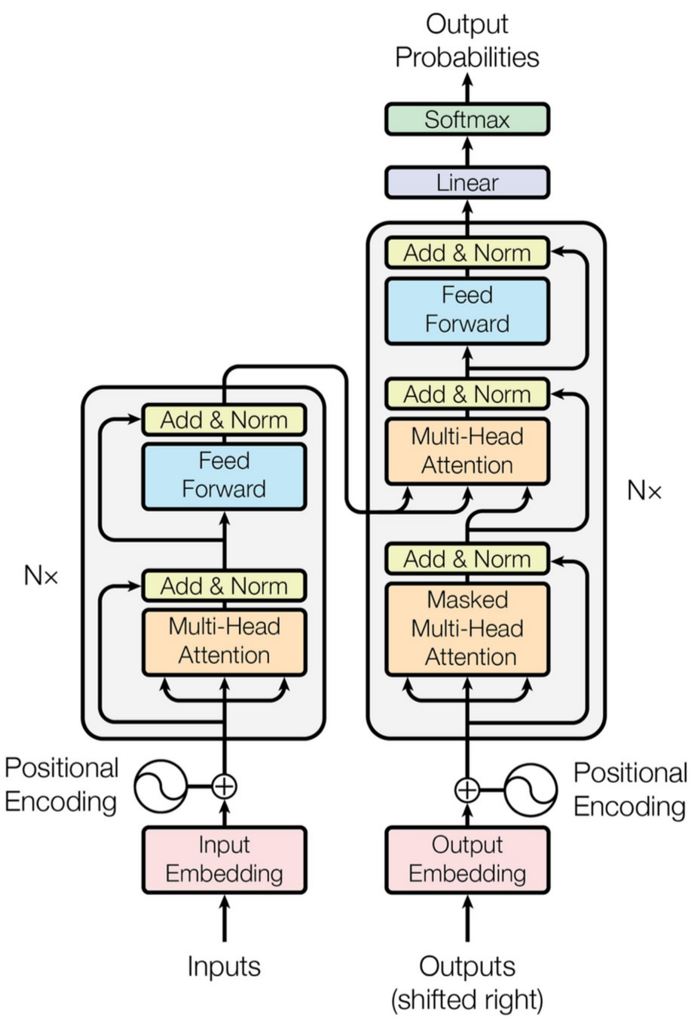



Figure 1: The Transformer - model architecture.

Image Credit: https://medium.com/@paruchurisaiteja15/attention-networks-3ab23176bb88

# Large Language Models

- Essentially a massive transformer based statistical model language
  - Trained on massive amounts of data
  - Has billions or trillions of dimensions
  - Takes massive compute power to train, usually using specialized hardware and parallel processing
- Transformers introduced
  - Self-attention
  - Parallel processing
  - Long-range dependency modeling
  - Efficient scaling

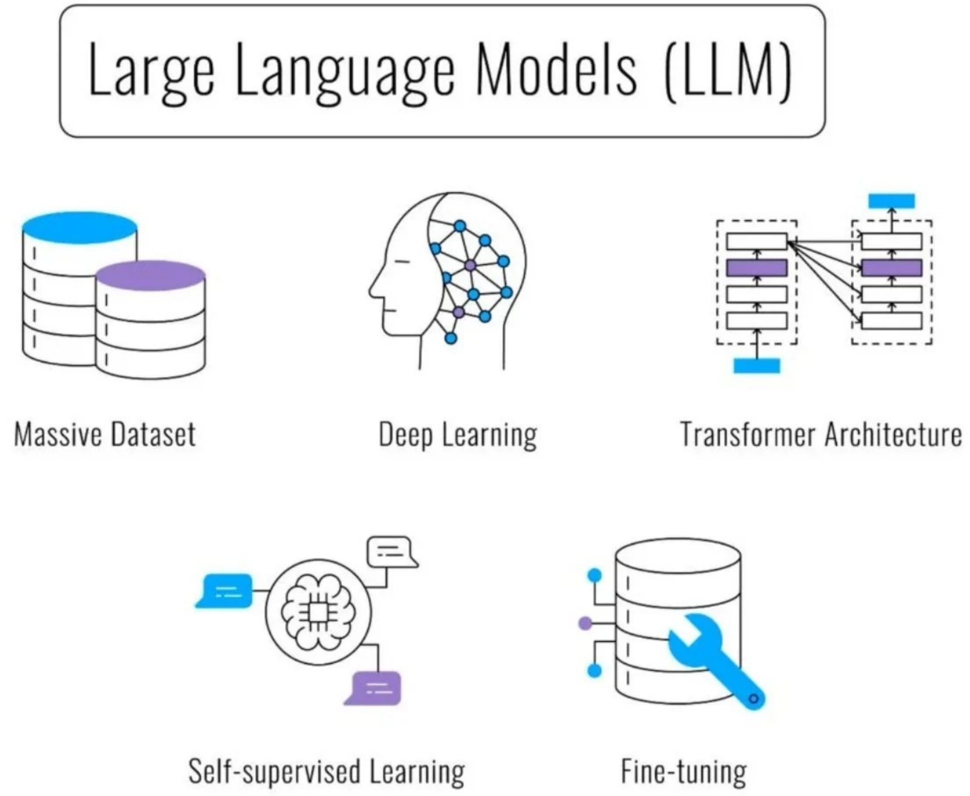
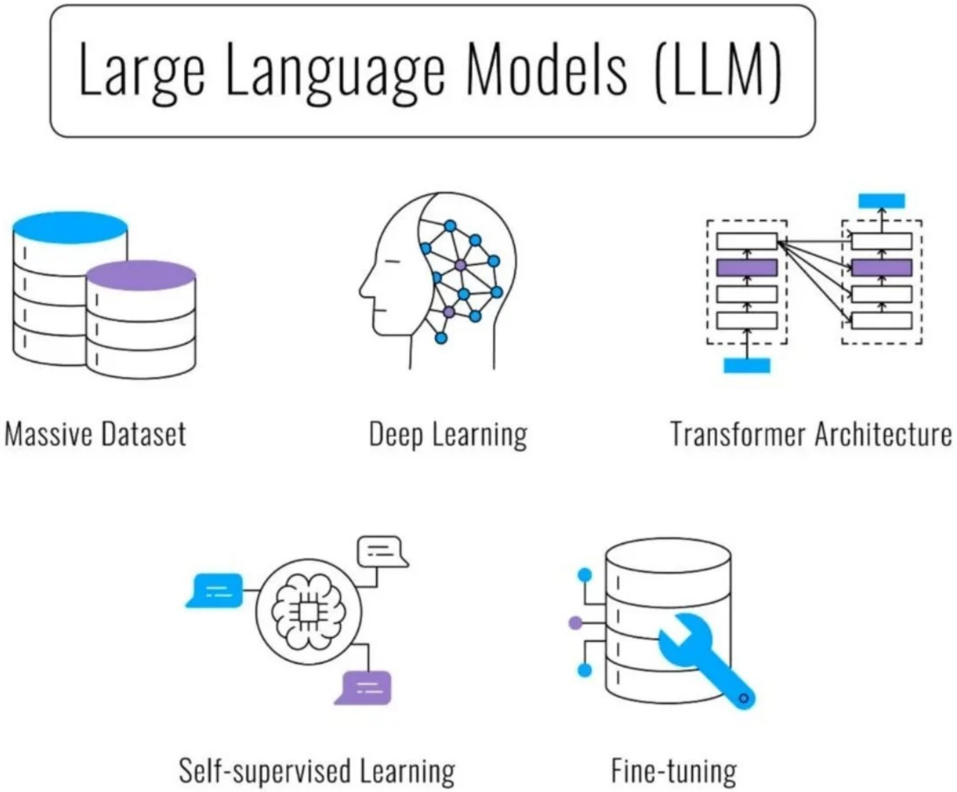



Image Credit: https://em360tech.com/tech-articles/what-large-language-model-llm-definition-examples-use-cases

# Large Language Models

- An LLM is
  - A transformer-based neural network trained on massive text data to predict and generate language
- It combines
  - Transformer architecture
  - Large-scale pretraining
  - Next-token prediction objective
  - Fine-tuning and alignment
- It is the basis for AI code assistants
  - Since code is just text, LLMs are good at developing large scale code statistical models

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# Multi-modal Training

- Multi-modal means training a model on multiple types of data
  - Text (including source code), images, audio, or video
  - Enables processing cross different types of information
  - Use cases
    - Image captioning
    - Visual question answering
    - Text-to-image generation
    - Document understanding (text + layout + images)
    - Voice assistants
    - Video understanding
    - AI Coding assistants
- How it works (conceptually)
  - Separate encoders for different modalities
  - Shared embedding space
  - Cross-modal attention mechanisms
  - Joint training objective

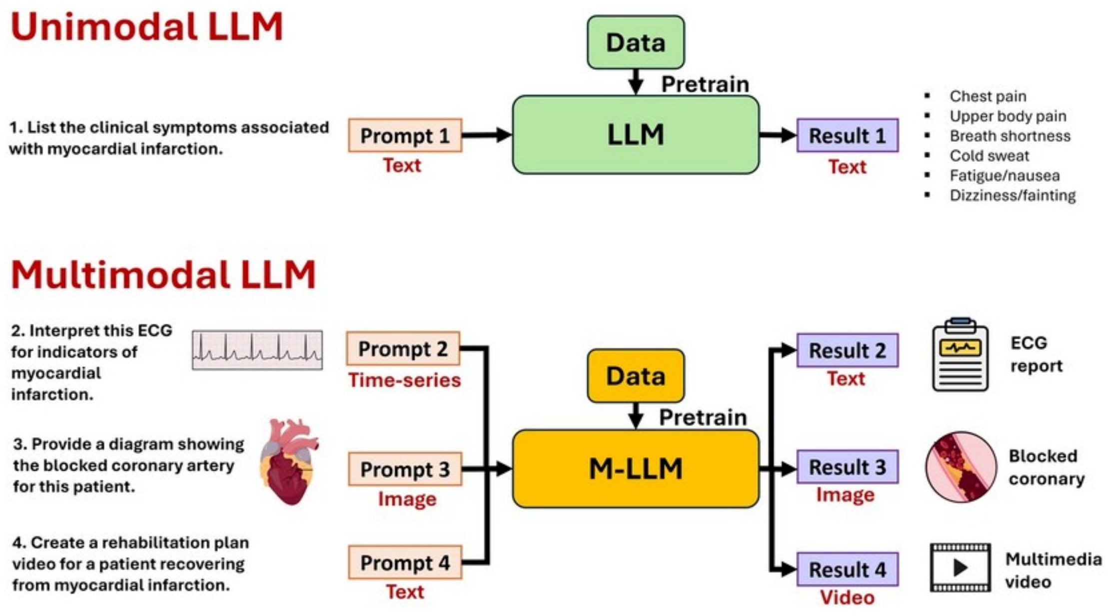



Image Credit: https://www.researchgate.net/figure/Unimodal-large-language-model-LLM-and-multimodal-LLM-M-LLM-examples-ECG_fig1_384331053

# Foundation Models

- Large-scale model trained on broad multi-modal data
  - Can be adapted to many downstream tasks
- Basic characteristics
  - Trained on massive, diverse datasets
  - General-purpose rather than task-specific
  - Can be fine-tuned to adapt the model to different tasks or context (like coding)
  - Serves as a base (“foundation”) for many applications
- The term LLM is often used when people are referring to a foundation model

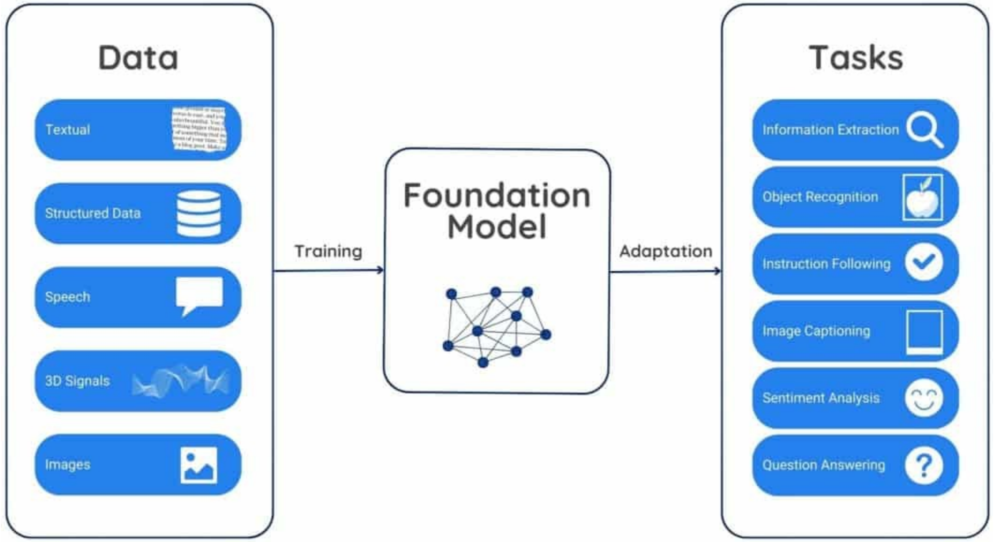



Image Credit: https://viso.ai/deep-learning/foundation-models/

# Traditional ML vs LLM vs Foundation Model

| Dimension | Traditional ML | Large Language Model (LLM) | Foundation Model |
|---|---|---|---|
| **Scope** | Task-specific | Language-focused | General-purpose |
| **Training Data** | Labeled, narrow dataset | Massive text corpora | Massive, broad datasets (text, images, audio, etc.) |
| **Training Objective** | Optimize for a specific task | Next-token prediction | Broad pretraining objective (often self-supervised) |
| **Model Size** | Small to medium | Billions+ parameters | Billions to trillions of parameters |
| **Architecture** | Many types (trees, SVM, CNN, etc.) | Transformer-based | Usually transformer-based (or diffusion for vision) |
| **Adaptability** | Must retrain per task | Prompting + fine-tuning | Prompting, fine-tuning, adapters, RAG |
| **Generalization** | Limited outside trained task | Strong within language domain | Cross-domain generalization |
| **Data Requirement for New Task** | Large labeled dataset | Few-shot / instruction | Few-shot / instruction / multimodal |
| **Example Use** | Spam classifier | ChatGPT-style chatbot | GPT-4, Claude, Stable Diffusion |
| **Development Model** | Train → Deploy | Pretrain → Align → Deploy | Pretrain → Adapt → Build ecosystem |

# The Generative AI Ecosystem

- Generative AI today is shaped by three major forces
  - Frontier model labs (training large foundation models)
  - Cloud & infrastructure providers (compute + distribution)
  - Open-source communities (democratization + rapid innovation)
- The ecosystem is highly interconnected:
  - Labs build models
  - Cloud providers distribute them
  - Open-source drives experimentation and adoption

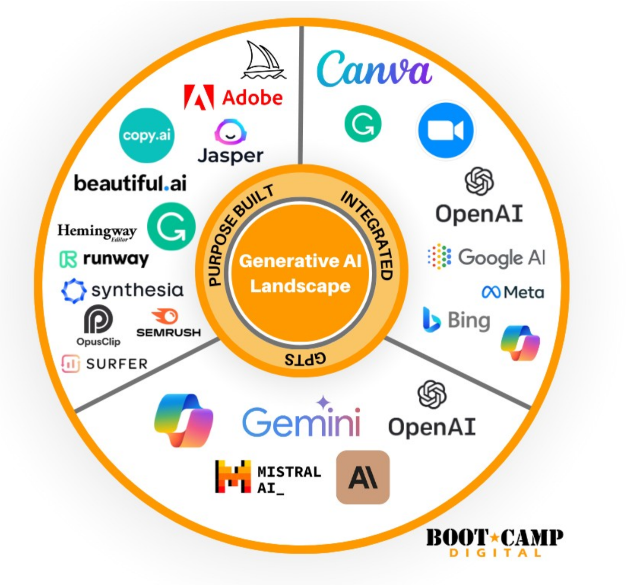



Image Credit: https://www.linkedin.com/pulse/generative-ai-landscape-ecosystem-bootcampdigital-ehdwc/

# The Generative AI Ecosystem

- Frontier model labs
  - Trains state-of-the-art foundation models
  - Operates at massive computational scale
  - Invests heavily in AI research and safety
  - Works with billions to trillions of parameters
- Pushes new capabilities
  - Reasoning, multimodality, agents, etc
- Requires enormous capital and specialized infrastructure

# The Generative AI Ecosystem

- Frontier labs operate where
  - Costs are extremely high
  - Technical uncertainty is high
  - Capabilities are advancing rapidly
- Examples
  - OpenAI
  - Anthropic
  - Google DeepMind
  - Meta (AI research division)
  - Mistral (emerging European lab)

# Why So Few Frontier Labs

- Training frontier models requires
  - Massive GPU clusters
  - Advanced distributed systems
  - Specialized ML researchers
  - Access to large-scale datasets
  - Billions of dollars in funding
- Only a small number of organizations can afford this.

# Why Frontier Labs Matter

- Set the pace of AI capability
- Define new benchmarks
- Influence regulation and safety standards
- Shape the global AI ecosystem
- Provide the foundation models others build upon

# Commercial Model Providers

- OpenAI
  - Pioneer of GPT series (GPT-3, GPT-4, GPT-4o)
  - Popularized the LLM product model (ChatGPT)
- Strong focus on
  - Multimodal models
  - Tool usage
  - Agents
- Deep partnership with Microsoft
  - Business model: API + enterprise licensing
  - Strategic role: Set early pace in scaling LLMs commercially.

# Commercial Model Providers

- Anthropic
  - Creator of Claude models
- Strong emphasis on
  - AI safety
  - Constitutional AI
  - Alignment research
- Competes directly with OpenAI in enterprise LLM space
  - Partnerships with Amazon (AWS Investment)
  - Strategic role: Safety-focused frontier model lab

# Commercial Model Providers

- Google DeepMind
  - Gemini family of models
- Long research history in
  - Transformers
  - Reinforcement learning
  - AlphaGo, AlphaFold
  - Strong multimodal capabilities
- Deep integration with Google products
  - Strategic role: Research powerhouse + platform integration.

# Commercial Model Providers

- Meta
  - LLaMA model family
  - Open-weight strategy
- Focus on
  - Open ecosystem
  - Developer adoption
- Major driver of open model competition
  - Strategic role: Accelerating open-weight ecosystem.

# Commercial Model Providers

- Microsoft
  - Infrastructure provider (Azure AI)
  - Strategic investor in OpenAI
- Embeds LLMs into
  - Office
  - GitHub Copilot
  - Enterprise stack
  - Focus on AI as enterprise productivity layer
- Strategic role: Distribution + enterprise integration.

# Commercial Model Providers

- Amazon
  - Bedrock platform (multi-model access)
  - Anthropic partnership
  - AWS infrastructure dominance
- Focus on
  - Enterprise AI infrastructure
  - Model hosting and orchestration
- Strategic role: AI as cloud service.

Amazon Bedrock

# Commercial Model Providers

- Mistral
  - European frontier lab
  - High-performance open-weight models
- Strong in
  - Efficient architectures
  - MoE (Mixture of Experts)
- Positioned as independent alternative to US giants
  - Strategic role: Efficient, competitive open-weight models.

# Open-Source Ecosystem

- Hugging Face
  - Central hub for open models
- Model hosting + sharing platform
  - Transformers library
  - Datasets + evaluation tools
  - Community-driven experimentation
- Strategic role: GitHub of AI models

# Open-Source Ecosystem

- LLaMA Family (Meta)
  - Open-weight large language models
  - Spawned thousands of derivatives
  - Enabled startups to fine-tune powerful models
  - Backbone of many open-source AI systems
- Impact: Democratized high-quality LLMs.

# Open-Source Ecosystem

- Stable Diffusion
  - Open-weight diffusion model for image generation
- Triggered explosion in
  - AI art tools
  - Custom fine-tuning
  - Local model usage
- Major shift from proprietary-only image generation
  - Impact: Brought generative image creative tools to the masses.

# Open-Source Ecosystem

- Falcon, Mistral (open models)
  - Falcon (UAE-based TII)
  - Mistral models (European lab)
- Competing in performance with frontier labs
  - Strong in efficient training and deployment
- Impact: Expanding global AI competition.

# Industry Dominance

- The generative AI landscape is:
  - Oligopolistic at the frontier (few companies can train trillion-parameter models)
  - Highly competitive in open-weight space
  - Deeply dependent on cloud infrastructure
  - Increasingly shaped by safety, regulation, and geopolitics
- The coding assistants and AI support tools are all based on generative models

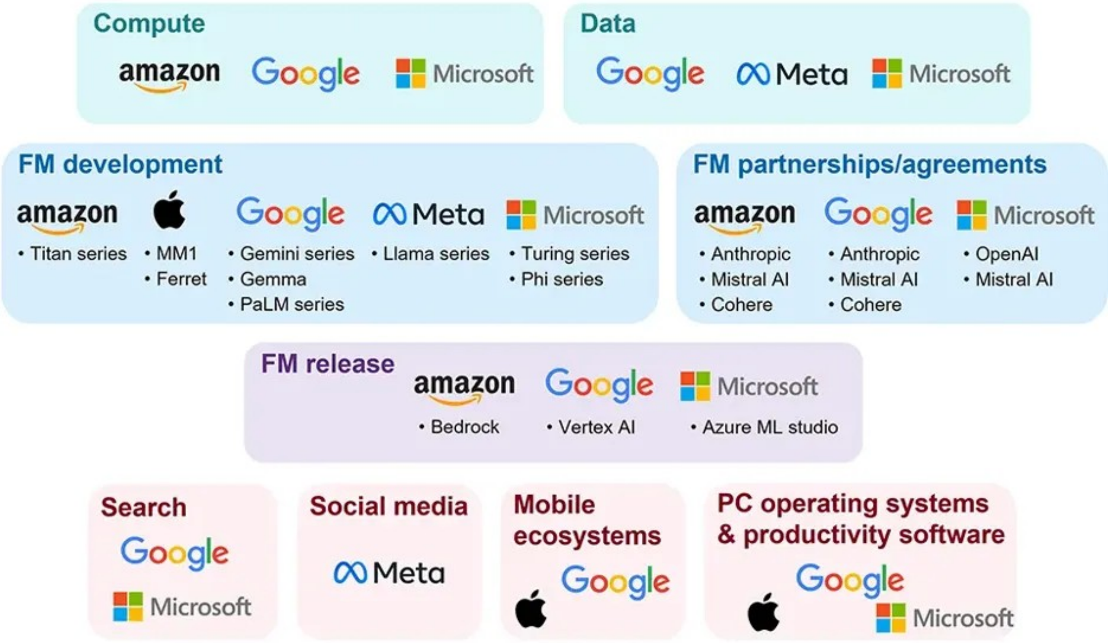



Image Credit: https://medium.com/a-i-society/ai-top-of-mind-for-5-13-24-oligopoly-f1644c48e879

# Current Trends

- Smaller and more capable models
  - Rapid improvement in
    - 3B–15B parameter models (B= billion)
    - Distilled frontier models
    - Quantized production-ready variants
    - Sparse / mixture-of-experts (MoE) inference optimization
  - Performance gap between
    - 70B+ models
    - Well-tuned 7B–13B models
  - is shrinking for many enterprise tasks

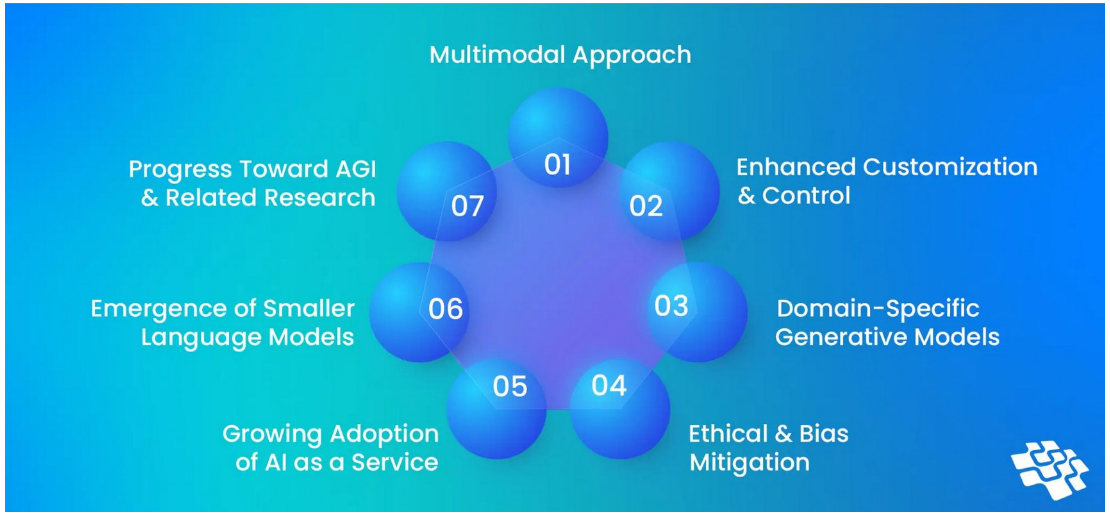



Image Credit: https://nextgeninvent.com/blogs/future-of-generative-ai-promises/

# Current Trends

- Smaller and more capable models
  - Within 1–2 years, many orgs will
    - Run high-quality models on-prem or VPC
    - Fine-tune smaller models cheaply with LoRA
    - Avoid relying exclusively on frontier APIs
  - This impacts
    - Infrastructure strategy
    - Cost modeling
    - Vendor lock-in decisions
    - Integration of AI based software development tools in development infrastructure

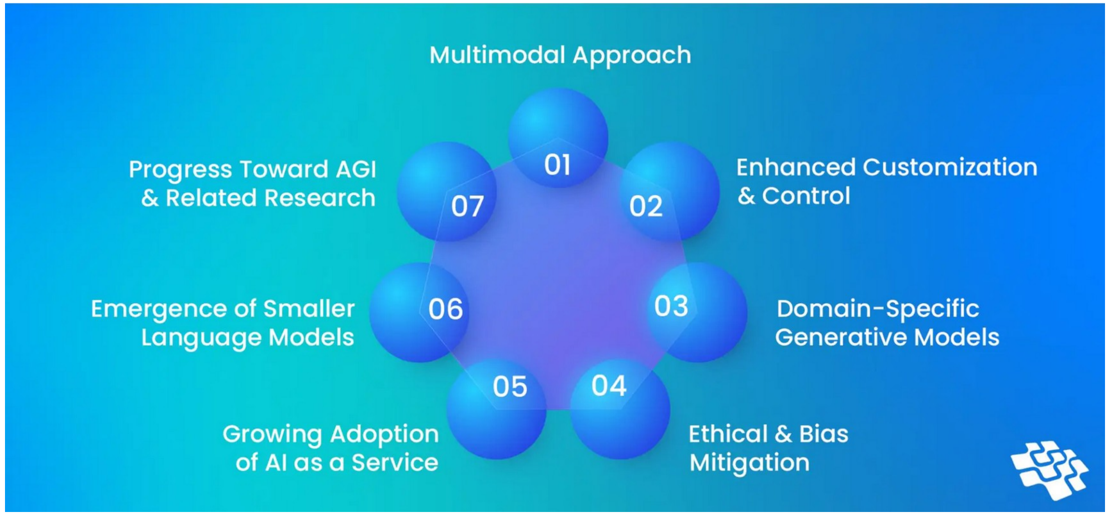
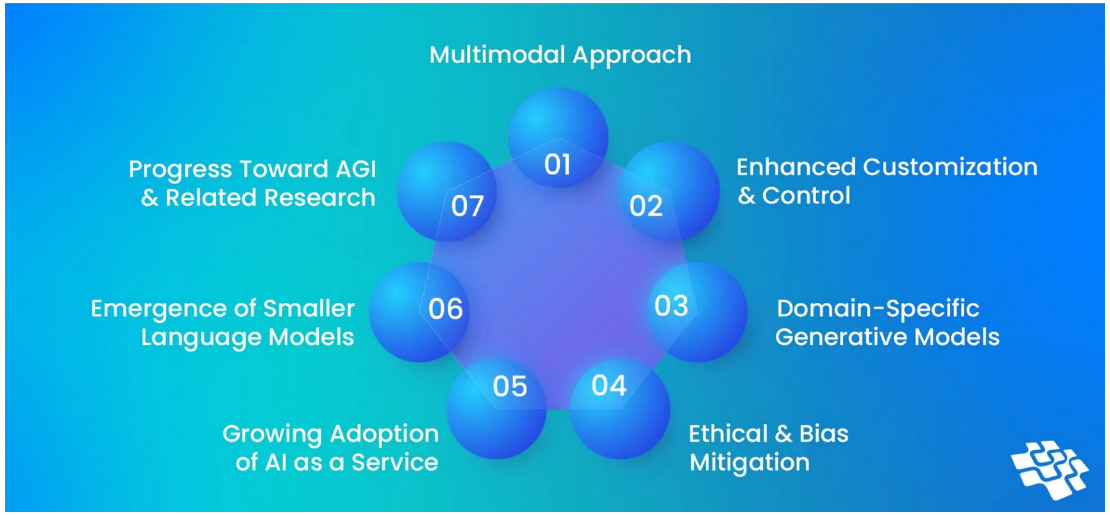



Image Credit: https://nextgeninvent.com/blogs/future-of-generative-ai-promises/

# Current Trends

- Native multimodal models
  - Multimodal models are becoming:
    - Native (not stitched pipelines)
    - Better at cross-modal reasoning
    - More reliable in production
  - In the near future:
    - Video understanding APIs will mature
    - Speech-to-reasoning-to-action pipelines will be standardized
    - *Diagram-to-code workflows will improve significantly*

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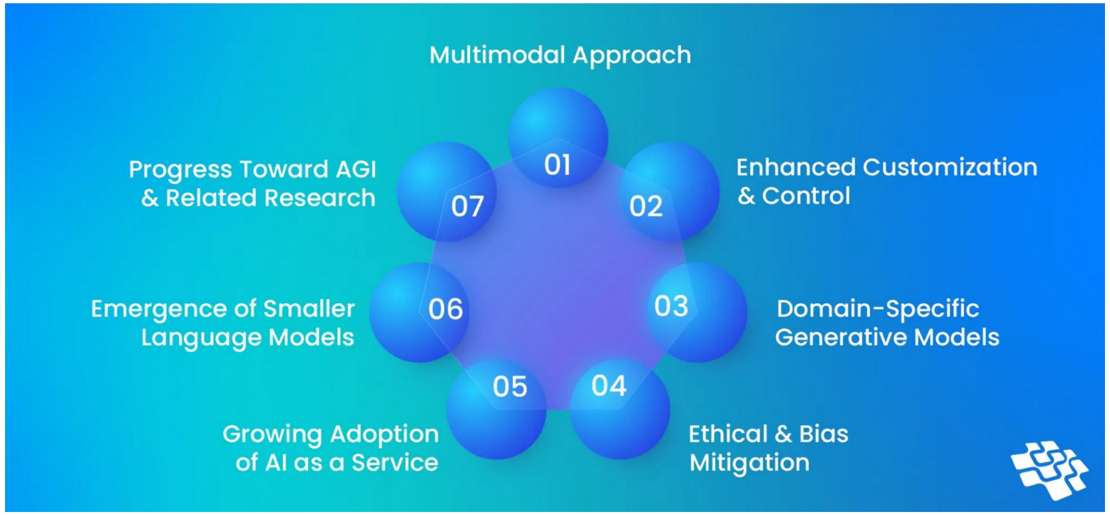
# Current Trends

- Structured output for tool first consumpton
  - LLMs are shifting from “Generate text” to “Generate structured actions”
  - Trends in LLMs
    - Function calling as default
    - JSON schema enforcement
    - Tool routing layers
    - Deterministic execution wrappers
  - In the near future:
    - Schema-constrained decoding to improve
    - Fewer malformed JSON outputs
    - Native validator-integrated decoding
    - LLMs become planners and not just executors

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# Current Trends

- Agentic systems with guardrails
  - Agents are real  but chaotic without control
  - In the near future:
    - Constrained agent frameworks
    - Finite-state agent controllers
    - Graph-based execution planners
    - Hybrid symbolic + LLM systems
    - Not fully autonomous AGI-style agents
    - Controlled orchestration systems
  - What will mature
    - Multi-step reasoning pipelines
    - Long-horizon task execution with supervision
    - Retry and self-correction loops
    - Observability tooling for agents

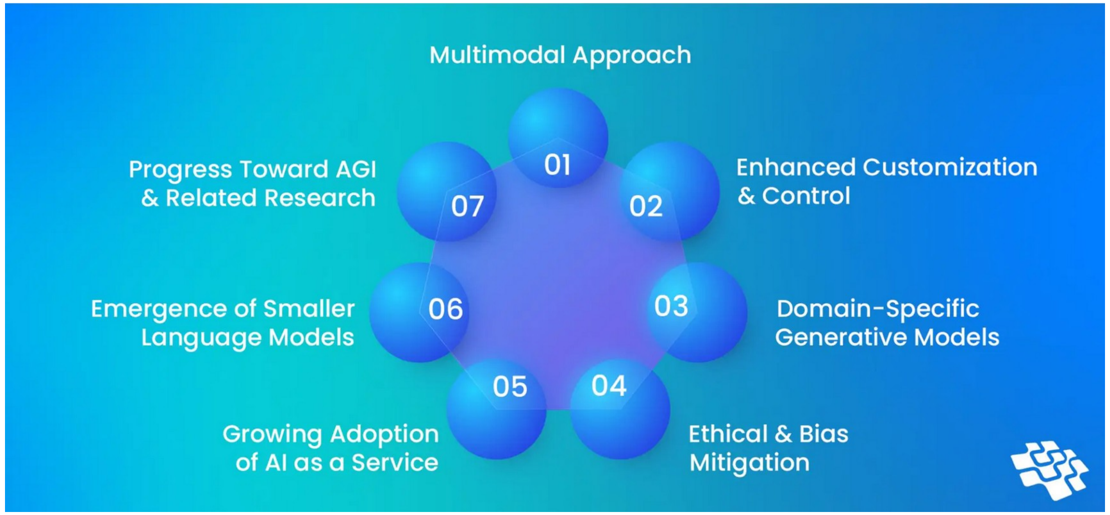



Image Credit: https://nextgeninvent.com/blogs/future-of-generative-ai-promises/

# Current Trends

- AI observability and evaluation tooling
  - Right now evaluation is primitive
  - In the near future
    - Standardized LLM evaluation frameworks
    - Drift detection
    - Prompt regression testing
    - Hallucination scoring tools
    - Behavioral fingerprinting
- This will probably be mandatory in enterprise applications

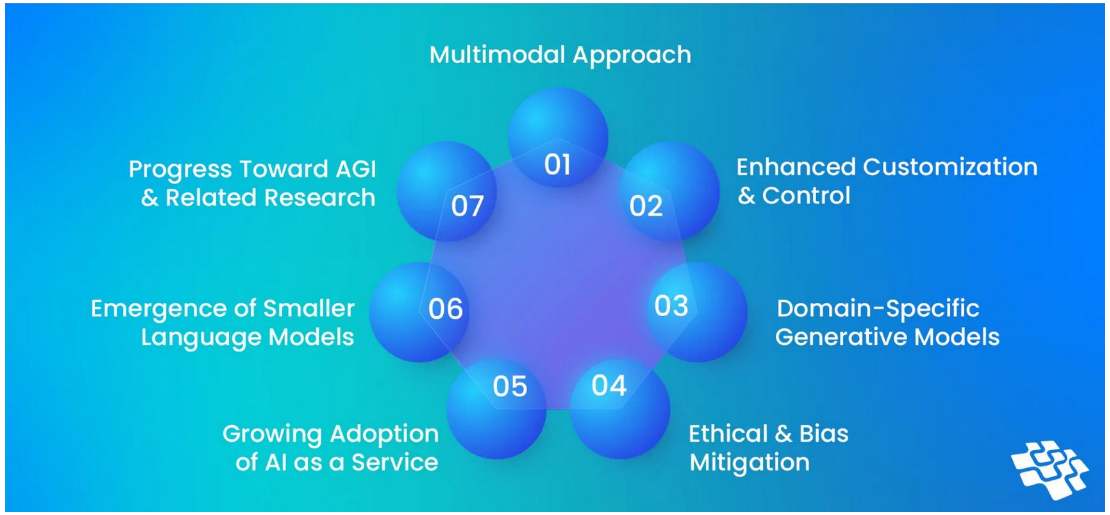



Image Credit: https://nextgeninvent.com/blogs/future-of-generative-ai-promises/

# Current Trends

- AI observability and evaluation tooling
- This will require
  - LLM test suites
  - Prompt versioning
  - Output diff tooling
  - Structured evaluation datasets
- AI will be treated like production software, not magic

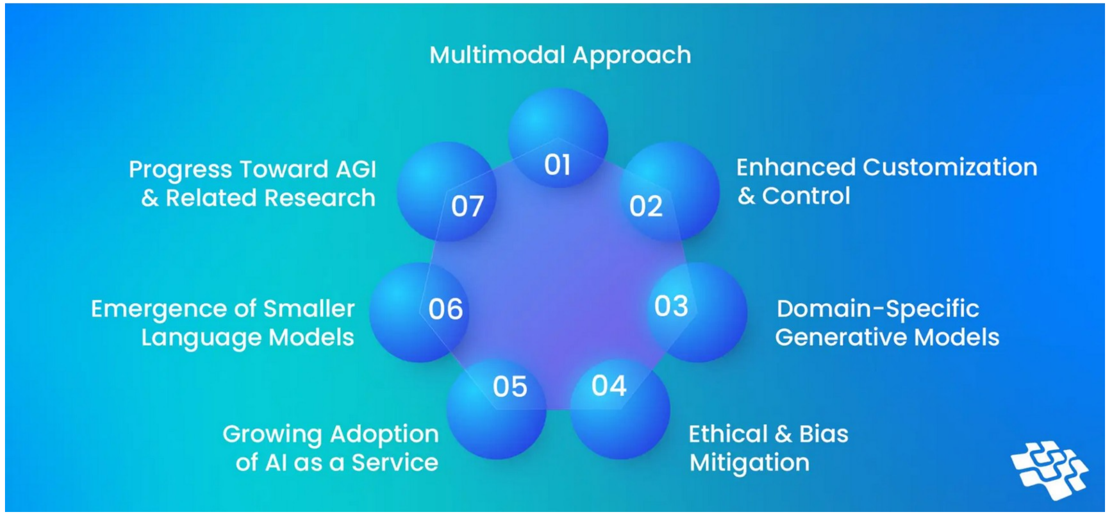



Image Credit: https://nextgeninvent.com/blogs/future-of-generative-ai-promises/

# Long Term Trends

- Long term trends (10-20 years)
  - Quantum computing and ML
  - Neuromorphic hardware
    - Hardware that mimics brain-like structures
  - Bio-computing, DNA-based computing
    - Most likely use: data storage
    - Silicon-biological hybrid systems
  - Photonic optical computing
    - Perform matrix multiplication using light interference
    - Near-zero energy cost per operation.
    - Extremely high parallelism

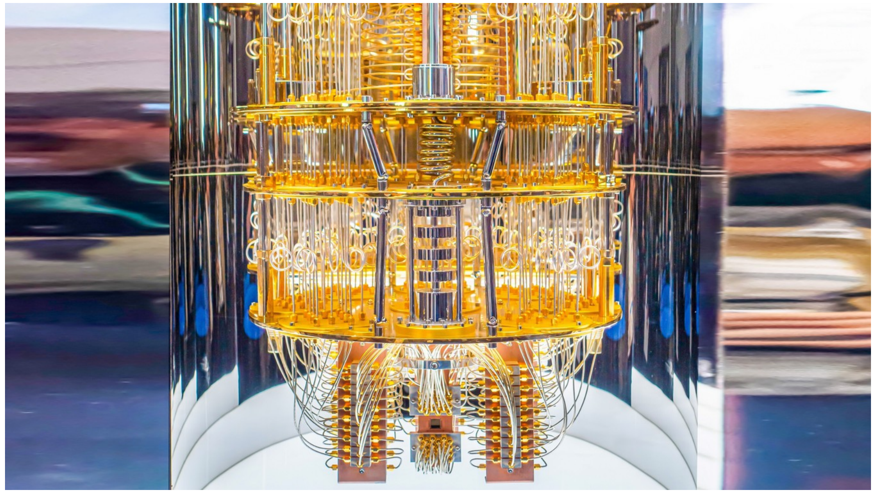
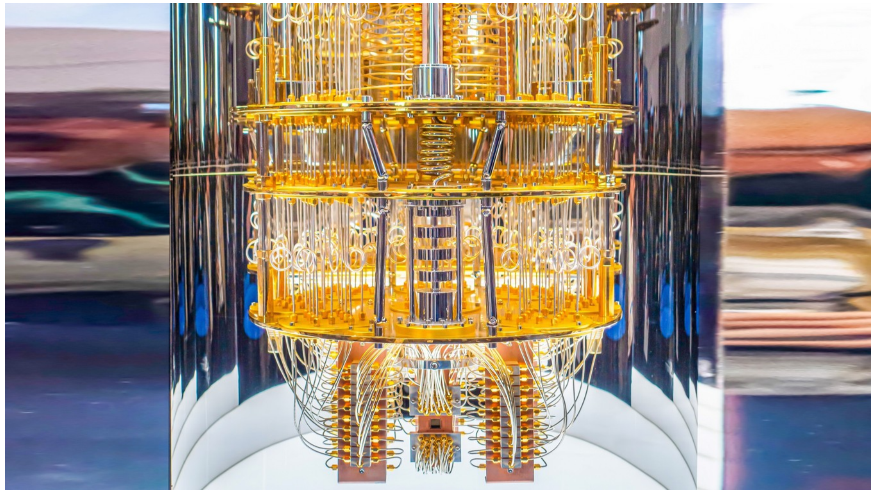


Image Credit: https://www.tomshardware.com/tech-industry/quantum-computing

# Long Term Trends

- Long term trends (10-20 years)
  - Collective AI systems
    - Consist of specialized AI swarms
    - Operate as distributed reasoning networks
    - Use consensus-based decision-making



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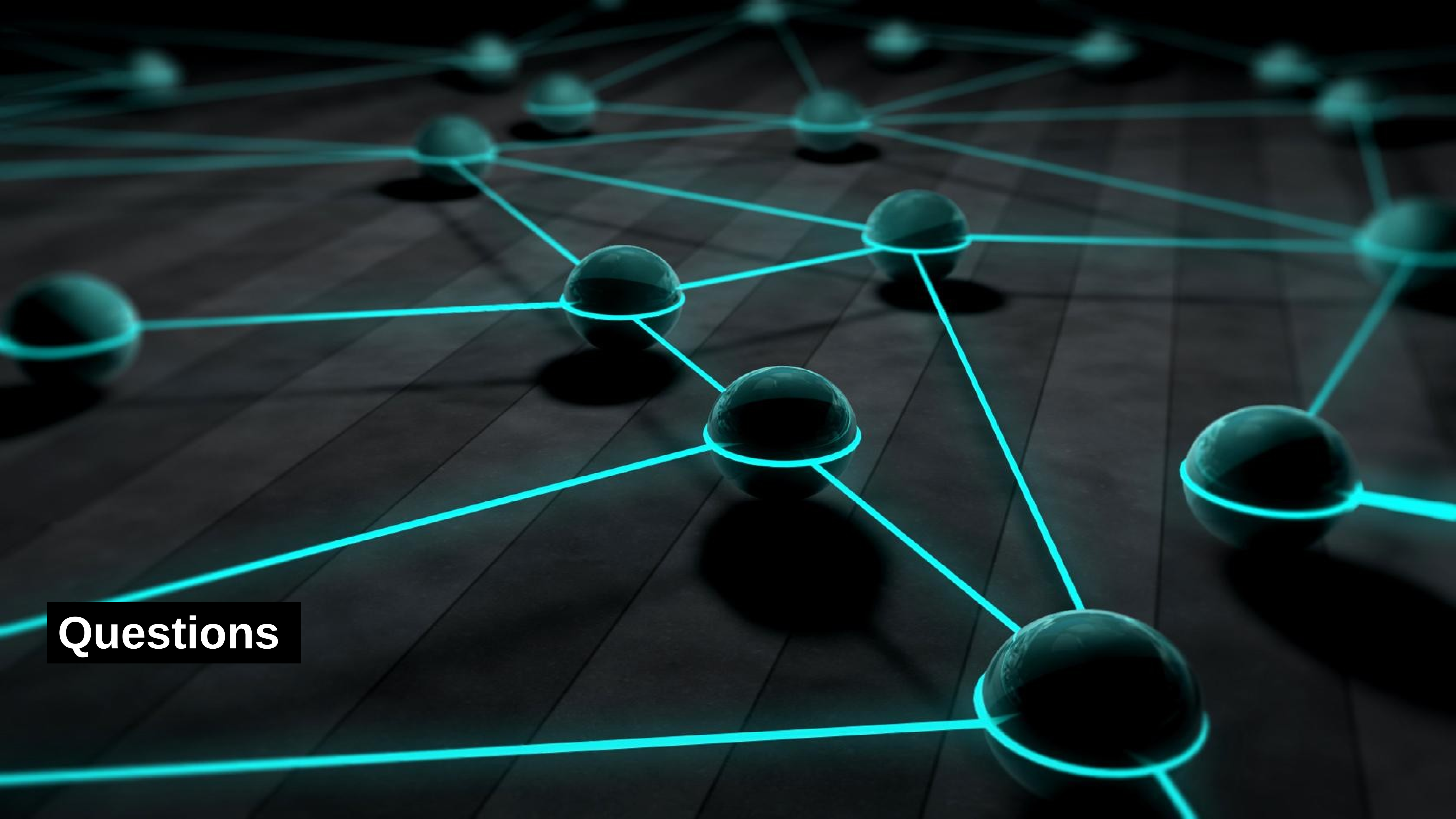

# Questions